

CLOUD COMPUTING AND VIRTUALISATION

① Objectives:

→ To understand the fundamentals of cloud computing, virtualization and hypervision, to launch and observe the lifecycle of an Amazon EC2 instance, and to apply AWS IAM users, groups, roles and policies so that access is granted following the principles of least privilege.

② Brief Theory:

→ Cloud computing delivers computing resources - servers, storage, networks and software - over the internet on demand, charged by usage. It is built on virtualization, where a hypervisor splits one physical server into several isolated virtual machines. AWS exposes this through EC2, which provides virtual servers launched from an AMI, and secure access to those servers through IAM.

③ PART A : Theory Questions

①. Cloud computing is the delivery of computing resources such as servers, storage, networking, and software over the internet on a pay-as-you-use basis. Two characteristics are on-demand self-service and rapid scalability / elasticity.

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Model	Provider gives	User Manages	Example
IaaS	Virtual infrastructure	OS, application, data	Amazon EC2

PaaS	Ready development platform	Application and data	AWS Elastic Beanstalk
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SaaS	Complete software	Data and setting	Gmail
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③. Virtualization creates virtual versions of physical resources, allowing multiple isolated virtual machines to run on one physical machine. It is important because it improves resource utilisation, reduce cost, provides isolation, and enables easy scaling.

④. Virtual Machine Physical Machine

Software-based computer

Actual hardware

Multiple VMs can share hardware

Uses its own hardware

Easy to create, copy & delete Requires physical setup

Lower cost and flexible Higher cost & maintenance

⑤. A hypervisor is software that creates and manages virtual machines.

- Type 1: Runs directly on hardware; faster and commonly used in data centers.

- Type 2: Runs on top of a host operating system; mainly used for testing and learning.

⑥ Amazon EC2 is an AWS service that provides resizable virtual servers called instances.

Three features are:

- Different instance types for different workloads.
- Elastic scaling using Auto-Scaling.
- Pay-as-you-go pricing.

⑦ An Amazon Machine Image (AMI) is a template containing the operating system and required software for an EC2 instance. It is required because it provides the software image from which an instance is launched and allows repeatable deployments.

⑧ AWS IAM manages user authentication and permissions for AWS resources. IAM should be used instead of sharing root credentials because it provides individual accounts, limited permissions, better security.

⑤ PART B: Conceptual Understanding

① (a) Sign in to AWS and open EC2.

(b) Select an AMI.

(c) Choose an instance.

(d) Create/Select a key pair.

(e) Configure the security group and network.

(f) Configure storage.

(g) Review and launch the instance.

(h) Observe the instance changing from pending $\rightarrow$ running.

②.

- Stopping: shuts down the instance while preserving EBS data.
- Restarting: Reboots the instance without deleting data or changing its instance identify.
- Terminating: Permanently deletes the instance. The action cannot be undone.

③.

- IAM user: An identify for a person or application.
- IAM group: A collection of users with common permission.
- IAM Role: Provides temporary permission without permanent credentials, e.g., an EC2 role accessing S3.
- IAM Policy: A JSON document defining allowed or denied actions on resource.

④.

- (a) No, it is insecure, prevents individual tracking, and gives excessive access.
- (b) Creates 50 individual IAM user, place them in one group, and attach a suitable policy to the group.
- (c) Give read-only EC2 permission. Student should not be allowed to terminate or delete resource.
- (d) It reduce the risk of accidental damage risk, misuse, unauthorized access, and unexpected AWS charges.

PART C: Practical Activity

Procedure

- Open the AWS EC2 dashboard.
- Selected launch instance and Amazon linux 2023 AMI.
- Selected a suitable instance type.
- Created and downloaded a key pair.
- Created a security group allowing SSH from my IP.
- Kept the default EBS storage and launched the instance.
- Observed Pending → Running → Stopping → Stopped.
- Created an IAM group with Amazon EC2 Read Only Access and test the permission.

Important Observation

Parameter	Value
Instance type	t2.micro
AMI	Amazon linux 2023
Region	ap-south-1
Key Pair	ec2-lab-key
Root Volume	8 GiB gp3 EBS
Initial State	Pending
Final State	stopped
IAM user	student 1
Policy	AmazonEC2ReadOnlyAccess

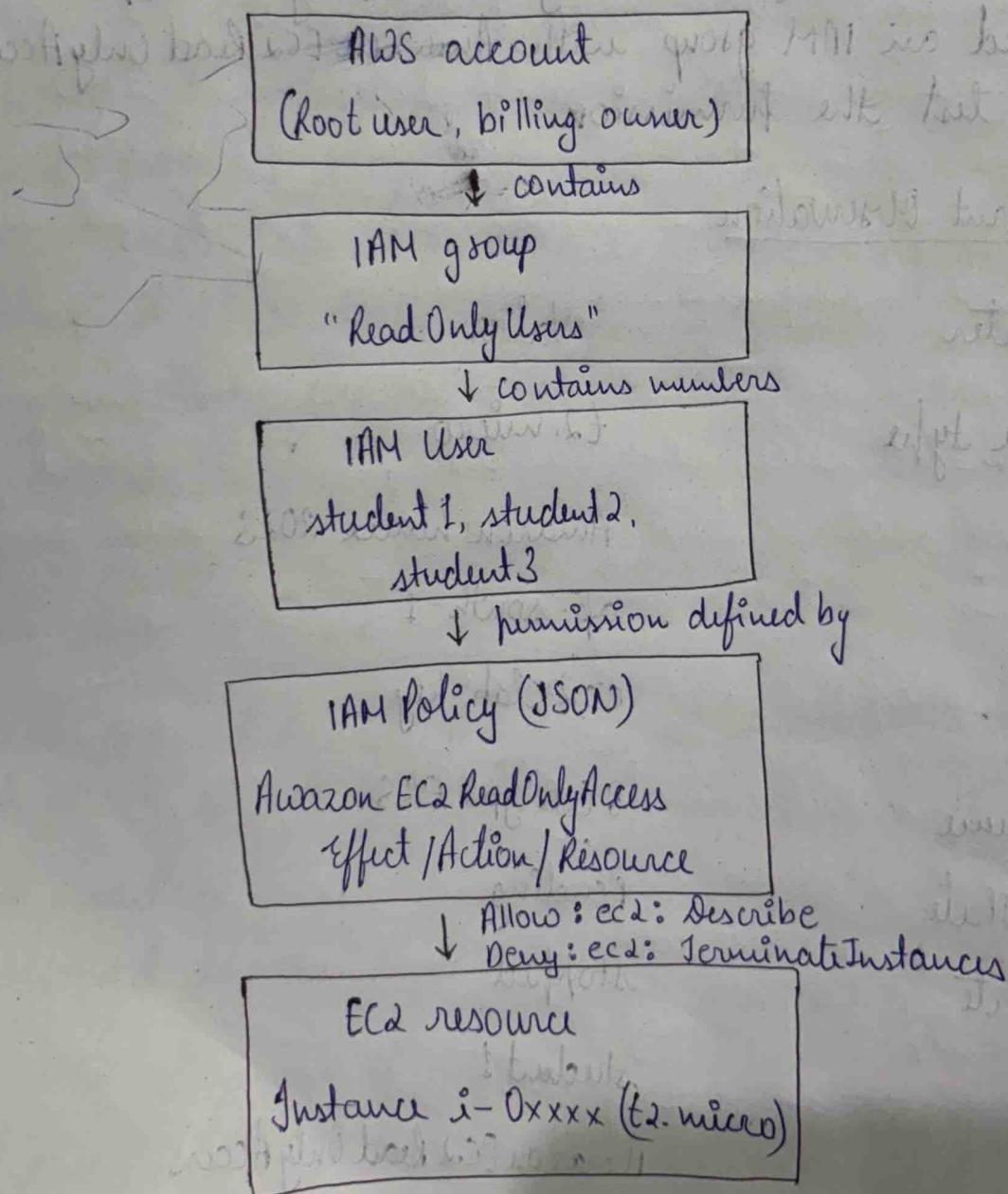
① State Transition

Pending → Running → Stopping → Stopped

② Why should students not get administrative permissions?

→ Administrative permissions could allow students to terminate instances, delete storage, create users, or launch expensive resources. Read-only reduces these risks and follows the principle of least privilege.

③ Diagram



② Conclusion

This assignment demonstrated the basis of cloud computing, virtualization, hypervisor, EC2, AMI and IAM. The EC2 lifecycle was observed from pending to running to stopped. IAM users, groups, and read-only policies were used to provide secure access while following the principle of least-privilege.

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